⚡ Zero45 – Cheat sheet for the game master

Page 1: preparation and explanation script (about 10 min) · Page 2: start, during the game, questions, problems. Black = explain, 💡 green = add this tip.

Before the start

1. Create the game, **timer stage 1 ≈ 30 min** (default 20 + explanation)
2. Team laptops: open the link with the PIN, **stop at the team name**; language via the **EN / DE** button bottom left
3. Host laptop: second tab, join as demo team “Game master”
4. **Start stage 1** (otherwise there is no Build tab), then explain in the demo team

Explanation script

1 Goal of the game — 1 min

👉 Overview at the very top

- **4 stages** up to 2045, each year calculated hour by hour
- **3 goals**, 0–5 points each: ⚡ supply (hours without power) · 🌿 climate (CO_2-free share *and* Mt CO_2 – **the worse one counts**) · 💶 costs (€/MWh)
- At most 60 points – **around 40 is very good**

2 The stage — ½ min

👉 yellow box on the host page

- Stage 1: 599 TWh, peak load 106 GW, coal ≤ 17 GW, CO_2 ≤ 70 Mt

💡 **Stage 2 is the big jump** (+35 % demand) – **what you build stays**: building ahead early pays off

3 Influence cards — 1½ min

👉 “Influence” tab

- **2 tokens**, 1 per card, each card once; more tokens for 2 build tokens each
- Left, policy: **works immediately** · right, technology: **only from the next stage**
- **Hydrogen only with a card**: demonstration programme = own electrolysis, grid = imports

💡 Hydrogen for 2040 → take the card **in stage 2 at the latest**

💡 Planning lots of onshore wind / ground-mounted PV → **acceptance early** (half the protest risk)

4 Build — 3 min

👉 “Build” tab: cards on the left, traffic lights on the right

- **18 tokens** at €10 bn; the card shows GW per token, ⓘ details; **leftovers expire**; credit up to +4 (+20 % costs)
- Keep or retire old power plants – **retiring is final**
- Protest → technology blocked for one stage; in stage 4 −2 points
- Traffic lights = firm capacity vs. peak load. **At the start both are red** (75 vs. 106 GW)

💡 **Build broadly** – spreads the protest risk

💡 **Gas turbines**: 15.6 GW per token, run only in scarce hours → fastest way to green traffic lights, can later be converted to hydrogen

💡 **Coal** always runs at ≥ 15 % – costs CO_2 even when you hardly need it

💡 **Old wind turbines** go offline (onshore only 30 % left in 2045) – replace them!

5 Simulator — 1½ min

👉 “Simulator” tab, run it once

- **3 test runs per stage**, normal weather, no chance
- Shows points, the **next point threshold**, a 💡 tip and the comparison with the last test – the real evaluation looks the same

💡 **Test early** (around half time), not only at the end

6 Chance — 1 min

👉 just tell them

- The same for everyone: weather (normal 60 / difficult 30 / extreme 10 %), demand ±3 %, disaster (20 → 50 %)
- Protests: for each team individually, depending on what it builds

💡 **Plan a buffer**: keep the traffic lights green, just green in the simulator is often not enough

💡 **Imports are no safe cushion**: in a Europe-wide lull they are missing, and they do not count as CO_2-free

7 Submitting and teamwork — ½ min

👉 “Mark as ready” button, timer at the top

- “Mark as ready” – can be changed until the end; notice 2 min before; without it the current state counts

💡 **Split the tasks**: one person builds, one tests, one keeps an eye on cards and time

💡 After each stage read the 💡 **tip** in the “Results” tab

Let's go

1. Teams enter their name → “Join” → straight into stage 1
2. Check the time, adjust with “−5 / −2 / +2 / +5 min” if needed; simulations with −/+ (per team or for all)
3. **Remove the demo team from the team list before stage 1 is evaluated** – otherwise it appears in the ranking

During the game

- **New stage:** show the yellow box, read out the warning line:
  - **Stage 1:** Coal at most 17 GW. Use influence tokens early.
  - **Stage 2:** Big jump: demand +35 %, peak load +37 GW – and only 6 GW of coal left.
  - **Stage 3:** Coal and oil are gone, reserve only 4 GW. Many old wind turbines go offline.
  - **Stage 4:** Climate-neutral: 5 climate points only with ≥ 97.0 % CO_2-free and ≤ 8 Mt CO_2.
- **After the evaluation:** read out the events line, put the interim standings (ranking + bars) on the projector briefly
- **At the end:** “End game & award ceremony” → podium

Typical questions

- **“The simulator looked fine – why gaps anyway?”** Different weather or a disaster; the simulator only calculates a normal year.
- **“Where are my leftover tokens?”** Build tokens expire, influence tokens stay.
- **“Why does the hydrogen turbine hardly count for the traffic light?”** Only as far as the hydrogen lasts for 2 weeks.
- **“Can I undo ‘Mark as ready’?”** Yes, until the stage is closed.

If something goes wrong

- **Page hangs:** reload (F5) – the team is stored in the browser. **Do not switch browsers**, no private window.
- **Laptops cannot reach the host:** usually the Wi-Fi blocks connections between devices → test the /join address with a second device beforehand (fixes: README "Network tips"); fallback: phone hotspot or own router.
- **Host crashed:** restart → “Continue running games”, teams reload. If an evaluation was running: everyone marks as ready again.
- **Evaluation takes a while:** normal with many teams, the display shows the progress.